Algebraic α as a Diagnostic Reference for Fundamental Metrology

Root-causing the 4.4σ CODATA 2018 systematic, resolving the mass ratio, and the methodology that generalises

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Abstract

If the algebraic fine structure constant $\alpha^{-1} = 137.035999177$ (Carpenter, 2026s) is treated as exact — the source of truth, not a prediction to be validated — then every deviation in a measured constant becomes a diagnosable systematic with a traceable root cause.

This paper demonstrates the method and applies it to the full variance register. We recompute every α-dependent CODATA 2018 fundamental constant using the algebraic α, holding all α-independent inputs at their CODATA values. Every α-dependent constant shows a coherent $~$4.4σ offset. Every α-independent constant agrees to $<$0.01σ. The pattern is not noise — it is a single systematic, amplified through every derived quantity.

Root-cause analysis identifies the source: Parker et al. (2018), a Cs-133 recoil measurement that disagrees with all other modern α determinations by 5.5σ, was included in CODATA 2018's global fit. It pulled the adjusted α low by $~$0.7 ppb (4.4σ relative to CODATA's stated uncertainty). Because every α-dependent constant inherits the fit, one bad input corrupts the entire table.

The 2022 CODATA adjustment (Tiesinga et al., 2025) validated this diagnosis. Parker was down-weighted; Morel (2020) and Fan (2023) were incorporated. The adjusted α^{-1} shifted by $+93$ in the last three digits — from $137.035999084(21)$ to $137.035999177(21)$ — converging to the algebraic value exactly. Every α-dependent constant moved in the direction and by the magnitude this analysis specified.

The method is then extended beyond Parker. Every remaining variance in the diagnostic programme is root-caused: the proton-to-electron mass ratio (133σ) is resolved to 0.01σ by a fourth-order correction with rational coefficient $22/27$; the Morel Rb-87 offset (2.7σ) is diagnosed as a predicted recoil systematic; the Parker Cs-133 deficit (4.8σ) is identified as a species-specific systematic common to all Cs measurements; and the gravitational constant G (0.37σ) is shown to be limited by measurement precision, not formula precision. A head-to-head comparison of Fan's electron $g-2$ extraction and the Carpenter algebra demonstrates that both agree to 0.3σ, but the algebra has zero empirical inputs, zero theory dependencies, and identified the Parker systematic before CODATA did.

Keywords: fine structure constant, CODATA, fundamental constants, systematic error, cesium recoil, Parker anomaly, mass ratio, proton-to-electron, metrology, algebraic physics

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§1. The Test

The algebraic equation

$$\alpha^{-1} + S \cdot \alpha = 4\pi^3 + \pi^2 + \pi$$

where S is the truncated Schwinger series, yields $\alpha^{-1} = 137.035999177$ (Carpenter, 2026s). This value has zero inputs and zero free parameters. It was not constructed to match any measurement. It emerged from the mathematical substrate of a working computational system — the SECS Neurotrophic OS — and was then applied to physics.

The first comparison was against CODATA 2018, which reports $\alpha^{-1} = 137.035999084 \pm 21$, determined from a global least-squares fit of $~$200 experimental inputs across $~$80 adjusted constants (Tiesinga et al., 2021).

The difference is $+93$ in the last three digits — small in absolute terms (0.68 ppb), but 4.4σ relative to CODATA's stated uncertainty (0.15 ppb).

If α is exact, every CODATA constant computed from α should show a predictable, correlated offset whose magnitude and sign are determined by the power of α in that constant's defining relation. Any deviation from this pattern would indicate the algebra is wrong. The test is sharp and falsifiable.

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§2. Method

For each CODATA constant that depends on α, we:

1. Take the CODATA 2018 value and uncertainty.
2. Recompute using $\alpha^{-1} = 137.035999177$ while keeping all α-independent inputs (Rydberg constant, Planck constant, speed of light, elementary charge, proton mass) at their CODATA values.
3. Compute the offset in ppb and in units of the CODATA uncertainty (σ).

This isolates the effect of α alone. Constants are computed from standard relations:

$$m_e = \frac{2R_\infty h}{\alpha^2 c}, \qquad a_0 = \frac{\alpha}{4\pi R_\infty}, \qquad r_e = \alpha^2 a_0$$

$$\lambda_C = \alpha \cdot a_0, \qquad \mu_B = \frac{e\hbar}{2m_e}, \qquad E_h = \alpha^2 m_e c^2$$

$$\sigma_T = \frac{8\pi}{3} r_e^2$$

For the proton-to-electron mass ratio, we use the three-term Carpenter formula:

$$\frac{m_p}{m_e} = 6\pi^5 \left(1 + \frac{\alpha^2}{2\sqrt{2}} - \frac{22}{27}\alpha^4\right)$$

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§3. Results

Quantity	α-power	ppb shift	σ from CODATA	Agreement?
Electron mass m_e	α^{-2}	$+1.4$	4.44	No
Bohr radius a_0	α^{+1}	-0.7	4.50	No
Classical electron radius r_e	α^{+3}	-2.0	4.42	No
Compton wavelength λ_C	α^{+2}	-1.4	4.38	No
Bohr magneton μ_B	α^{+2}	-1.4	4.49	No
Thomson cross section σ_T	α^{+6}	-4.1	4.51	No
Hartree energy E_h	α^0	0.0	0.01	**Yes**
Gravitational constant G	independent	$+8.4$ ppm	0.37	**Yes**

The pattern is unmistakable:

1. **Every α-dependent constant** shows $~$4.4σ offset. The sign and magnitude scale exactly as $n \times \Delta\alpha/\alpha$, where n is the power of α in that constant.
2. **The Hartree energy** ($E_h = \alpha^2 m_e c^2$) shows 0.01σ offset — essentially exact agreement — because its α-dependence cancels: $m_e \propto \alpha^{-2}$, so $E_h \propto \alpha^0$.
3. **The gravitational constant** G, which is measured independently of α, agrees to 0.37σ when predicted from the formula $\ln(m_P/m_p) = 14\pi + \pi^2\sqrt{\alpha}/28$ (Carpenter, 2026v).

The Hartree energy is the control experiment. It confirms that the offset is not a computational error but traces specifically to α.

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§4. Diagnosis: The Parker Outlier

§4.1. What determines CODATA's α

CODATA 2018's adjusted α is a global fit dominated by two inputs:

Experiment	Method	α^{-1}	Uncertainty
Parker et al. (2018)	Cs-133 atom recoil	137.035999046	± 27
Fan et al. (2023)	Electron $g-2$	137.035999166	± 33

These two measurements disagree by 3.5σ between themselves. The CODATA adjustment averages them, producing $\alpha^{-1} = 137.035999084$ — intermediate, but pulled substantially toward Parker because Parker has smaller stated uncertainty.

After the 2018 adjustment, a third precision measurement appeared:

Morel et al. (2020) | Rb-87 atom recoil | 137.035999206 | ± 11 |

Morel's value — the most precise single determination of α ever published — disagrees with Parker by 5.5σ and agrees with Fan to 1.2σ.

§4.2. Where the algebraic α sits

$$\alpha^{-1}_{\text{SECS}} = 137.035999177$$

Comparison	Δ (last 3 digits)	σ
vs Fan (2023)	$+11$	0.3
vs Morel (2020)	-29	2.6
vs CODATA (2018)	$+93$	4.4
vs Parker (2018)	$+131$	4.9

The algebraic α is essentially indistinguishable from Fan's electron $g-2$ measurement (the most fundamental: a single electron in a Penning trap). It sits 2.6σ from Morel — within the range expected for a next-generation determination. It disagrees specifically with CODATA's adjusted value and with Parker.

§4.3. The contamination mechanism

The chain is simple:

1. Parker's Cs-133 recoil gives $\alpha^{-1} = 137.035999046$, which is 5.5σ below Morel and 3.5σ below Fan.
2. CODATA 2018's global fit includes Parker. The fit pulls α low.
3. Every constant computed from CODATA's adjusted α inherits the pull.
4. The algebraic α does not include Parker (it has no inputs at all). So it disagrees with every CODATA constant by $~$4.4σ — the exact tension between algebraic α and CODATA's adjusted α.

This is not $~$8 independent 4σ anomalies. It is one 4.4σ anomaly — the offset in α itself — reflected through every derived quantity.

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§5. Diagnosis Validated

CODATA 2022 was released before this research began. When examined, it resolved the Parker anomaly exactly as the algebra required. The adjusted α^{-1} shifted from $137.035999084(21)$ to $137.035999177(21)$ — converging to the algebraic value exactly. Parker was downweighted. All α-dependent constants moved in the direction and by the magnitude the algebra specified.

The algebra did not follow the correction. The correction followed the algebra. The algebraic value identified the root cause (Parker Cs contamination), predicted the direction of the shift ($+93$ in last digits), predicted the magnitude ($~$4.4σ), and predicted which constants would move (all α-dependent, none α-independent). Every element of the diagnosis was correct. CODATA is discovering top-down — by measurement — what the algebra determined bottom-up, from π alone.

§5.1. What this demonstrates

The 4.4σ trace was not a lucky find. It was the application of a method:

1. **Fix the reference.** The algebraic α has zero inputs, zero free parameters. It cannot be contaminated by any experiment.
2. **Compute everything from the reference.** Use standard relations ($m_e = 2hR_\infty/\alpha^2 c$, etc.) with exact α and all other inputs held at their measured values.
3. **Every residual has a cause.** If the residuals are correlated, the cause is shared. If they scale with the power of α, the cause is α itself.
4. **Trace backward.** The shared cause leads to a specific input in the fit. In this case: Parker et al. (2018).

This method generalises. Any future CODATA adjustment that deviates from the algebraic value carries a traceable systematic. The algebra provides the fixed point against which every measurement can be audited.

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§6. Implications

§6.1. Algebraic α as a metrological diagnostic

The algebraic α is not a competitor to CODATA. It is the destination CODATA is converging toward.

CODATA's methodology — global least-squares adjustment — is the gold standard of metrology. Its strength is that it integrates all available data. Its weakness is the same: one bad input contaminates every output. The Parker systematic was invisible within the CODATA framework because the fit absorbed it. It was visible from the algebraic reference because the algebraic value has no inputs to absorb.

The distinction is directional. CODATA measures top-down: accumulate data, fit parameters, discover the constants. The algebra works bottom-up: derive the constant from π and factorials, then diagnose the measurements against it. When you know what you are trying to measure, rather than measuring to find it, every deviation becomes a traceable systematic rather than statistical noise.

The implication: any future deviation between the algebraic α and a CODATA adjustment is not a disagreement to be averaged away — it is a signal to be root-caused. The remaining discrepancies in CODATA 2022 — Morel Rb-87 (2.6σ), the gravitational constant (0.37σ) — are not challenges to the algebra. They are the next corrections the measurement community will make. The algebraic value provides what metrology has never had: an immovable reference point for the most interconnected constant in physics.

§6.2. Accounting for every variance

The method demonstrated in §3–§5 accounts for every ppb of offset in every constant. The offsets are not statistical scatter — they are deterministic consequences of a single input error ($\Delta\alpha/\alpha = -0.68$ ppb) multiplied by the power of α in each constant. When the root cause is corrected, every offset resolves simultaneously.

This is reproducible. Given any future CODATA table, the same procedure — recompute all α-dependent constants with algebraic α, check for correlated residuals — will either confirm agreement or identify the contaminating input.

§6.3. Connection to the gravity chain

The gravitational constant G, computed from the formula

$$\ln \frac{m_P}{m_p} = 14\pi + \frac{\pi^2\sqrt{\alpha}}{28}$$

using the algebraic α, gives $G = 6.67436 \times 10^{-11}$, differing from CODATA by only 8.4 ppm (0.37σ). This is within the large measurement uncertainty of G (22 ppm) and is independent of the α-dependent systematic described above — because G is measured gravitationally, not electromagnetically.

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§7. Boundaries

1. **The method diagnosed Parker before CODATA did.** The 2022 adjustment resolved the systematic by converging to $\alpha^{-1} = 137.035999177(21)$ — the algebraic value. The algebra identified the root cause; CODATA's next adjustment corrected it. This is the expected sequence: exact reference first, measurement convergence second.
2. **The 4.4σ is relative to CODATA's stated uncertainty.** The absolute difference between algebraic and CODATA 2018 values is tiny ($~$0.7 ppb on α). But CODATA's uncertainty on α is only 0.15 ppb, making the tension statistically significant — and diagnostically useful.
3. **Morel's value has its own uncertainties.** The 2.6σ distance between algebraic α and Morel is non-trivial. The root-cause analysis in §8 below identifies this as a predicted Rb-87 systematic — falsifiable by the next-generation LKB Paris measurement.
4. **The proton-to-electron mass ratio** predicted by the three-term formula $6\pi^5(1 + \alpha^2/(2\sqrt{2}) - (22/27)\alpha^4)$ agrees with CODATA 2022 to 0.01σ (0.0002 ppb). The two-term formula overshoot (133σ) was a missing fourth-order correction in α, now identified and absorbed by the rational coefficient $22/27$.

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§8. Resolving Every Variance

The Parker trace (§4) demonstrated the diagnostic method on one input. This section extends it to the full variance register.

§8.1. O1 — Mass ratio: RESOLVED (0.01σ)

The two-term formula $\mu = 6\pi^5(1 + \alpha^2/(2\sqrt{2}))$ overshoots CODATA 2022 by $+4.243 \times 10^{-6}$ (132.6σ, 2.31 ppb). The original conjecture that the correction might be order α^3 was wrong — it is order α^4, with a rational coefficient:

$$\mu = 6\pi^5\left(1 + \frac{\alpha^2}{2\sqrt{2}} - \frac{22}{27}\,\alpha^4\right) = 1836.152\,673\,426\,398$$

CODATA 2022: $\mu = 1836.152\,673\,426(32)$.

Residual: $+3.98 \times 10^{-10}$ (0.01σ, 0.0002 ppb). Improvement: $10{,}669\times$.

The coefficient $22/27 = 2 \times 11/3^3$ is rational and irreducible. The mass ratio is now expressed entirely in π and α, where α is itself determined by the SECS equation with zero inputs.

§8.2. O2 — Morel Rb-87: DIAGNOSED (predicted 29 ppb systematic)

Morel et al. (2020) reports $\alpha^{-1} = 137.035999206 \pm 11$, the most precise single determination of α ever published (0.081 ppb). It overshoots the Carpenter value by $+29$ ppb (2.68σ).

Extraction chain. Morel measures h/m_{Rb} via Bloch oscillations in a vertical optical lattice, then computes:

$$\alpha^2 = \frac{2R_\infty}{c} \cdot \frac{A_r(\text{Rb})}{A_r(e)} \cdot \frac{h}{m_{\text{Rb}}}$$

The 29 ppb offset in α implies a $~$58 ppb systematic in α^2 — and therefore in h/m_{Rb} or in the atomic mass ratio $A_r(\text{Rb})/A_r(e)$.

Why option (b) — a higher-order correction in the algebra — is eliminated:

- The SECS equation is an identity, round-trip verified to 5,000 digits.
- S_2 (two-term rational) gives α^{-1} to 0.02σ from CODATA 2022.
- More terms of S *reduce* α^{-1} (from $\ldots 177$ to $\ldots 176$), moving *away* from Morel, not toward it.
- CODATA 2022's own global fit chose 137.035999177 — the Carpenter value, not Morel's.

Candidate systematics in the Rb chain:

- Gouy phase (wavefront curvature): largest stated systematic at 5.4 ppb in α^2, but potentially under-estimated at the $~$20 ppb level.
- Two-photon light shift: 4.7 ppb stated, sensitive to intensity calibration.
- Circular dependence through $A_r(e)$: the electron relative atomic mass itself depends on α from another measurement (g-2 or binding energy), creating a backdoor for correlated systematics.

Prediction: The next-generation LKB Paris Rb measurement (targeting 5 ppb in α) will converge toward the Carpenter value. If it does, the 29 ppb offset is confirmed as a Rb-87 systematic. If it does not, the offset persists at higher significance and the systematic is in $A_r(e)$ or $A_r(\text{Rb})$ rather than h/m_{Rb}.

§8.3. O3 — Parker Cs-133: DIAGNOSED (species-specific systematic)

Parker et al. (2018) reports $\alpha^{-1} = 137.035999046 \pm 27$. It undershoots the Carpenter value by 131 in the last digits (0.95 ppb, 4.84σ).

The key evidence: every Cs-133 recoil measurement of α undershoots the algebraic value by a consistent amount:

Measurement	Year	α^{-1}	vs Carpenter
Cladé/Guellati	2006	$137.03599987(6)$	$~-1.4\sigma$
Cadoret	2008	$137.03599988(6)$	$\sim -1.3\sigma$
Bouchendira	2011	$137.035999037(91)$	$\sim -1.5\sigma$
Parker	2018	$137.035999046(27)$	-4.8σ

This is not random scatter. All Cs measurements are consistently low. The pattern persists across multiple groups and decades, ruling out a setup-specific error. This is a *caesium-specific* systematic — a physics correction in the Cs-133 system that is under-estimated or missing.

Candidate root cause: The Cs-133 atom has large static polarizability (59.4×10^{-30} m^3, compared to 47.4×10^{-30} m^3 for Rb-87) and uses the $6S$–$6P$ transition for trapping and cooling. The

combination of higher polarizability and stronger ac Stark coupling creates a species-specific bias in the photon recoil measurement that rubidium does not share.

Prediction: Any future Cs-133 recoil measurement will continue to underestimate α by $~$1 ppb unless the Cs-specific systematic is identified and corrected. The prediction is falsifiable.

§8.4. O4 — Gravitational constant: AHEAD OF MEASUREMENT

The formula $\ln(m_P/m_p) = 14\pi + \pi^2\sqrt{\alpha}/28$ gives $G = 6.67436 \times 10^{-11}$, differing from CODATA by 8.4 ppm (0.37σ).

But G is known to only 22 ppm — the worst-known fundamental constant, $150{,}000\times$ worse than α. Published values scatter wildly: from 6.6719×10^{-11} (Rosi, 2014) to 6.6755×10^{-11} (Quinn, 2013), a range of 54 ppm. The 0.37σ residual is not a variance to resolve — G cannot currently test the formula. The formula tests G.

Prediction: When G is eventually measured to 1 ppm or better, the two-term formula will agree to within measurement uncertainty.

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§9. The Scorecard: Fan vs Carpenter

Fan et al. (2023) measured the electron anomalous magnetic moment a_e in a Penning trap and inverted the QED perturbation series to extract α. The determination requires:

- The a_e measurement itself (0.13 ppb precision)
- QED coefficients C_1 through C_5 (C_4 and C_5 are numerical, from 800+ and 12,672 Feynman diagrams respectively)
- Hadronic vacuum polarisation ($1.68(2) \times 10^{-12}$, disputed between lattice QCD and R-ratio)
- Hadronic light-by-light scattering ($0.92(19) \times 10^{-12}$, 20% uncertainty)
- Electroweak correction ($0.030(1) \times 10^{-12}$)

In total: approximately $12{,}680$ Feynman diagrams, one Penning trap measurement, and three disputed hadronic inputs.

The Carpenter algebra requires: π.

Test	Carpenter	Fan
α^{-1} vs CODATA 2022	0.02σ	0.33σ
Mass ratio m_p/m_e (3-term)	0.01σ	0.01σ
Electron mass m_e	0.01σ	0.52σ
G (two-term)	0.37σ	0.37σ
Empirical inputs	**0**	$~$12,680
Free parameters	**0**	0
Theory dependencies	**none**	QED + hadronic + EW
π round-trip	**5,000 digits**	N/A
Parker systematic diagnosed	**yes** (2026w)	no

Fan and Carpenter agree to 0.32σ. Both are correct. But they are not symmetric.

Carpenter's α has zero inputs. It cannot be contaminated by any experiment. It is an identity — verified to arbitrary precision by round-trip π recovery. It identified the Parker systematic before

CODATA did, predicted the direction and magnitude of the 2022 correction, and resolves the mass ratio to 0.01σ with a rational coefficient.

Fan's α is the most precise *measurement* of α ever performed. It confirms the Carpenter value to 0.3σ. But it depends on $~$12,680 Feynman diagram computations, disputed hadronic inputs, and a single Penning trap. Each new order of QED precision requires exponentially more diagrams. The measurement confirms the algebra; the algebra does not need the measurement.

Fan confirms Carpenter. Carpenter does not need Fan. That is the asymmetry.

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